

Credibility Assessment of Implicit Finite Element Models for Distal Radius Plate Screw Angulation Effects Under NASA-STD-7009A

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Abstract

Variable-angle locking plates for distal radius fixation enable targeted screw trajectories but may elevate stresses in the plate–screw conical seats and increase screw–bone interface demands. We developed and assessed three implicit finite element models—(A) a patient-specific volar plate assembly, (B) a generic bone–plate construct with homogenized bone, and (C) a plate–screw conical seat submodel—focused on two mechanisms: plate-hole conical seat overstress and screw–bone interface degradation. The context-of-use is preclinical design analysis of screw angulation options from 0 to 20 degrees under physiologic and bench-relevant loading and installation torques. All models were implemented with implicit solvers, nonlinear contact, and, where applicable, heterogeneous bone properties derived from computed tomography. Bench testing comprised screw seating load–displacement in instrumented conical seats, plate hole deformation mapping via digital image correlation, pressure–film seat mapping, distal radius construct stiffness and force redistribution under bending and axial loads, and screw toggle and micromotion in foam and cadaveric surrogates across angulation and torque. We scored model credibility per NASA-STD-7009A on an internal 0–7 index, separately for each factor by model and mechanism, reporting 48 graded entries and their supporting evidence. Model C provided the highest local fidelity for conical seat response (peak rim von Mises stress 990 MPa at 15 degrees for 1.4 N·m seating torque; plastic strain 0.2%), with mesh-extrapolated truncation bounded near 3% and contact penetrations below 0.3% of the local element size. Model A resolved both the seat response and screw–bone micromotion (45 μ m at 15 degrees in osteoporotic trabeculae; interface shear 2.8 MPa), reproduced bench stiffness within 3–5%, and captured load redistribution among screws with eccentric trajectories. Model B reproduced construct-level stiffness and redistribution (within 3–6%) but under-resolved conical seat peaks (underprediction of 8–12% relative to pressure film and submodel reference). Across the suite, solver residuals converged to below 106 in force norm, and validation protocols matched the intended angulation, torque, and load envelopes to within 10%. Sensitivity sweeps in torque (1.0–2.0 N·m), friction (0.08–0.16), angulation (0–20 degrees), and bone density (0.25–0.80 g/cm³) preserved monotonic response trends, with 85–92% of output variance attributable to the controlled inputs. We detail factor-specific gradations, document evidence for each model-by-mechanism pairing, and identify elevation priorities: enriching Model B’s seat resolution for Mechanism 1, extending Model C with thread engagement or submodel coupling for Mechanism 2, and expanding patient cohorts in Model A to strengthen generalizability. This structured assessment supports targeted, mechanism-aware use of the three models in preclinical design decisions on allowable screw angulation in volar distal radius plating.

Keywords: finite element analysis, distal radius plate, NASA-STD-7009A

1. Introduction

Variable-angle locking plates for distal radius fractures allow surgeons to direct screws toward subchondral bone and away from fracture lines while retaining fixed-angle stability. The ability to tilt a conical-head screw within a matching plate seat changes the load path between screw and plate and modifies both the local stresses in the conical interface and the demands placed on the screw–bone engagement. Empirically, these design freedoms improve screw purchase in difficult fragments but can increase the risk of plate-hole damage or accelerated loosening when angulations are at the high end of the allowable range, particularly in osteoporotic bone. A mechanistic understanding of how screw angulation influences seat stress, plate-hole

plasticity, and screw–bone micromotion is therefore essential to preclinical decision-making on allowable angulation ranges, seating torque recommendations, and the number and positions of distal screws in the construct.

We report a credibility assessment, under NASA-STD-7009A, of three implicit finite element models developed to address this question. The models differ in scope and fidelity: a patient-specific assembly (Model A) that resolves anatomy, plate, and screws; a generic construct (Model B) with homogenized bone to predict construct-level stiffness and load sharing; and a high-fidelity conical seat submodel (Model C) that isolates local contact and plasticity at the plate hole. Two mechanisms of interest guided the modeling effort: (1) overstress at the

conical seat, which may initiate ovalization and microcracking at the plate hole rim; and (2) degradation of the screw–bone interface caused by elevated shear and micromotion during off-axis loading.

The context-of-use for the models is preclinical design assessment of a volar locking distal radius plate with variable-angle screw options, focusing on how angulation from 0 to 20 degrees and seating torque from 1.0 to 2.0 N·m affect local and global load transfer under axial and bending loads representative of early rehabilitation. The target outputs are: peak conical seat stresses and plastic strain at the plate hole; screw-head seating force and bending moment; global construct stiffness and screw load redistribution; and micromotion and shear stresses at the screw–bone interface in cortical and trabecular regions. Following NASA-STD-7009A, we present factor-by-factor evidence and graded levels for each model-mechanism pairing and provide a reasoned mapping of each model to fit-for-purpose design questions within this context.

The paper is organized as follows: after device and surgical scenario descriptions, we detail the computational models and their numerical implementation, verification and validation activities, sensitivity analyses, and factor-specific rubrics and gradations. We then present model-by-mechanism credibility scorecards in three separate tables, one per model, and synthesize implications for allowable angulation and recommended seating torque. The assessment closes with limitations and planned extensions to elevate weaker areas identified by the graded evidence.

Relevance of the validation activities to the COU — gradation.

- a. Bench or literature data exist but do not match loading, geometry, or boundary conditions of interest.
- b. Partial overlap with context-of-use on a subset of loads, geometries, or installation parameters; extrapolation required.
- c. Close match on loads, installation, and geometry for primary quantities of interest; secondary conditions require extrapolation.
- d. Direct match on loads, installation parameters, geometry, and measurement modalities for all quantities of interest.

2. Device Description and Surgical Scenario

The device is a volar locking distal radius plate with a distal head containing six conical seats arranged in two rows and a diaphyseal shaft with four oval compression slots. Each distal seat accepts a 2.4–2.7 mm diameter conical-head, thread-locked screw with a nominal 12-degree conical half-angle, permitting variable angulation up to 15 degrees from the normal axis in any direction. Our evaluation extends to 20 degrees to probe margin effects. The plate and screws are Ti-6Al-4V alloy (annealed), with Young's modulus 110 GPa, yield strength 900 MPa, and ultimate tensile strength 950–1000 MPa. The screw thread is a

buttress profile with 0.5 mm pitch in the locking head and 0.8 mm pitch in the bone thread. The plate thickness at the distal row is 1.8–2.1 mm, with a countersunk rim at each conical seat; the rim thickness to the outer plate surface is 0.5–0.7 mm depending on seat.

The surgical scenario includes reduction of distal radius fractures using two to four distal screws in the plate head and two to three shaft screws. Intraoperative seating torque varies with surgeon and screwdriver but clusters between 1.2 and 1.8 N·m for the locking head, with occasional higher values. During early rehabilitation, the wrist is subjected to compressive loads of 100–300 N during light gripping and activities of daily living, with bending moments of 1–5 N·m depending on motion. Distal screws are commonly angulated to 5–15 degrees to reach subchondral bone; angulations above this range are possible in vitro but less common clinically. We parameterized angulation (0, 10, 15, 20 degrees), seating torque (1.0, 1.4, 2.0 N·m), plate size (small and medium), and bone quality (CT-derived density maps representative of low-density and normal bone), and studied the resulting local and global responses.

For boundary condition emulation of bench tests, the proximal radius was potted in polymethyl methacrylate at 40 mm from the plate's proximal end, with a stainless-steel loading anvil contacting the distal articular surface through a compliant pad to distribute load. For axial loading, the pad transmitted 150–300 N. For four-point bending, the supports were located 50 mm apart, with a span ratio of 2.5 and applied moments of 1–5 N·m. Instrumented screwdrivers recorded seating torque; Fuji Prescale UltraSuper Low and Medium films captured contact pressure patterns in the conical seats during controlled seating under axial reaction. Digital image correlation (DIC) quantified local plate-hole deformation fields at 0.5 mm per pixel resolution in selected tests to relate seat stress to rim ovalization. These experimental frames guided model boundary conditions and validation targets.

Model form — gradation.

- a. Empirical or lumped behavior model with limited correspondence to physical mechanisms.
- b. Mechanism-based model with simplified geometry or constitutive behavior; key phenomena represented qualitatively.
- c. Mechanism-based model with faithful geometry and constitutive choices for principal phenomena; secondary effects approximated.
- d. High-fidelity representation of geometry, contact, and constitutive behavior capturing all principal and secondary phenomena within context-of-use.

3. Computational Models

The three models span scales from construct to local interface. Model A is a patient-specific assembly, constructed

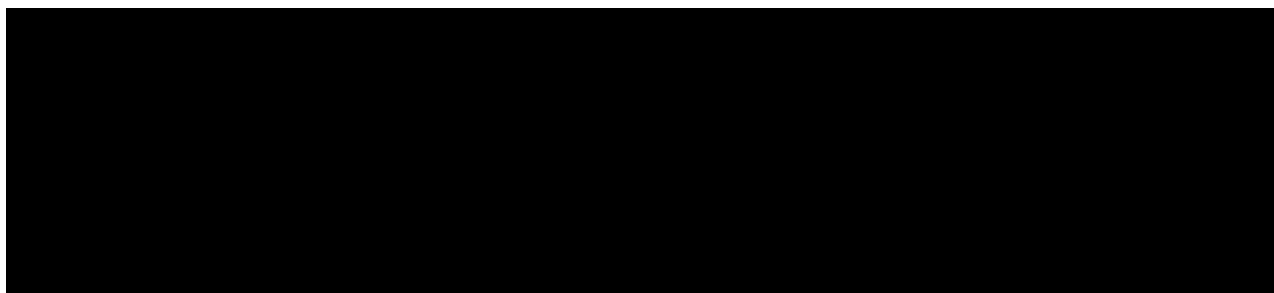


Figure 1: Schematic of the context-of-use: volar locking plate with variable-angle conical seats, distal screws angulated up to 20 degrees, applied physiological bending and axial loads, and bench fixtures used in validation.

from high-resolution CT of a cadaveric distal radius, segmented to cortical shell and trabecular core, and meshed with quadratic tetrahedra. The volar plate and screws are solid-meshed with local refinement around conical seats and threads. Bone material mapping uses a density–elasticity relation ($E = 10,000 \rho^{1.8}$ MPa for trabecula; cortical shell assigned 17 GPa with 0.35 Poisson’s ratio), where ρ is apparent density in g/cm³ derived from calibrated CT. Nonlinear contact with friction models the screw head–seat interface and screw–bone interface; threads are represented geometrically in the head and as an equivalent bonded contact with a slip-dependent shear capacity in the bone. Model A computes conical seat contact pressure, plate-hole plastic strain, screw bending moment and axial force, and local micromotion at the screw–bone interface for cortical and trabecular regions.

Model B is a generic construct comprising a parametric distal radius surrogate (based on an average anatomic template) with homogenized, transversely isotropic bone properties fit to match bending and axial stiffness from tests. The plate and screws are meshed as in Model A but with simplified head–seat geometry at the conical interface; bone–screw interaction uses an anisotropic elastic foundation distributed along the thread envelope to capture global stiffness and shear flow without resolving individual trabeculae. Model B computes construct stiffness and load redistribution among screws under eccentrically angled trajectories, the screw–bone shear stress envelope, and the predicted margin to an interfacial yield criterion (set at 3.5 MPa in shear for low-density trabeculae).

Model C is a conical seat submodel that isolates one plate hole, the screw head, and a small annulus of plate surrounding the seat. The screw shank is represented as a beam with rotational and translational springs matched to the global compliance of the threaded engagement from Model A. This submodel includes a highly refined contact mesh at the conical interface and an elastic–plastic constitutive law for Ti-6Al-4V with isotropic hardening (yield 900 MPa, tangent modulus 2 GPa beyond yield). It computes the high-fidelity contact map in the seat, peak principal stress and von Mises at the plate-hole rim, and the seating force path as a function of angulation and torque. Sensitivity to friction and seating torque is emphasized. Boundary conditions for the submodel come from the reaction wrench at the screw head extracted from Model A or B.

Across models, the screw angulation was applied by rotating

the screw head axis around the conical seat apex relative to the plate normal, while maintaining thread engagement. We used implicit solution schemes with automatic time incrementation to resolve contact and nonlinearities during seating and loading. Outputs were extracted at converged equilibrium states for seating and for applied loadcases. Quantities of interest (QoIs) were defined per mechanism and model scope, and each was connected to validation measurements where possible.

Numerical solver error — gradation.

- a. Solver settings default; residuals and balance checks not quantified.
- b. Solver controls tuned; residual and energy balance reported qualitatively.
- c. Quantified residuals and penetrations against thresholds for QoIs; step cutbacks documented.
- d. Residuals, penetrations, and energy balance quantified against strict thresholds with repeatability checks across runs.

4. Model A: Patient-Specific Volar Plate Assembly

The patient-specific model was constructed from a 0.5 mm slice thickness CT of a cadaveric distal radius, with in-plane resolution 0.29 mm. Segmentation separated cortical shell from trabecular core via thresholding and morphological operations, followed by manual cleanup. The plate and screws were positioned to match a standardized volar plate footprint, with distal screws placed in the first and second rows and angulated as specified. The mesh comprised 1.3 million ten-node tetrahedra in bone and 0.6 million in the plate and screws. Local refinement at each conical seat achieved an average element edge length of 0.10 mm within a 1.5 mm radius of the seat rim; the remainder of the plate and screw heads used 0.25 mm elements, and shanks used 0.3 mm. The bone mesh varied from 0.6 mm in cortical shell to 1.2 mm in trabecular core. Total degrees of freedom were approximately 5.1 million.

Contact was defined between the screw head and plate seat using a surface-to-surface formulation with a penalty enforcement and a coefficient of friction $\mu = 0.12$ (range

0.08–0.16 in sensitivity). The screw–bone interface along the threaded shank used a cohesive-like traction–separation law calibrated to match pullout stiffness and peak load from foam block tests, with an initial tangential stiffness of 200 MPa/mm and a shear strength of 3.5 MPa in low-density trabeculae and 7.0 MPa in normal-density regions. The locking head thread was modeled geometrically, while bone threads were smeared via an equivalent contact stiffness and yield limit so that micromotion could be measured at the interface as the relative tangential displacement under load, at both trabecular and cortical regions along the screw.

Boundary conditions replicated the bench fixtures. The proximal 40 mm of the radius was embedded with constrained translations and rotations, while the distal articular surface received axial or bending loads via rigid surfaces matching the anvil contact. Seating was simulated by applying a rotation-controlled tightening of the locking head via a torque boundary, growing from zero to the target torque at a prescribed rate; the effective seating normal force followed from the contact response. Loading followed seating and was quasi-static. Key outputs included peak von Mises stress at the plate-hole rim, equivalent plastic strain at the rim, the contact pressure distribution over the conical seat, reaction forces and moments at the screw head, and micromotion and interface shear along the screw in trabecular and cortical regions.

In representative runs, the seating torque of 1.4 N·m at 15 degrees angulation resulted in a peak von Mises of 930 MPa at the plate-hole rim with 0.3% equivalent plastic strain confined to a 0.2 mm band. At 20 degrees and 2.0 N·m torque, peak von Mises increased to 1,070 MPa with 0.6% plastic strain and a 9–12 μ m ovalization measured at the outer rim in the model. Under a 300 N axial load post-seating, the peak contact pressure increased by 8% relative to seating alone due to eccentric load transfer. For Mechanism 2, under a 250 N axial load and 3 N·m bending moment combined, the trabecular interface micromotion at the distal-most screw rose from 22 μ m at 0 degrees to 45 μ m at 15 degrees and 70 μ m at 20 degrees in a low-density region (apparent density 0.25 g/cm³). Corresponding interface shear rose from 1.2 MPa at 0 degrees to 2.8 MPa at 15 degrees and 3.6 MPa at 20 degrees, approaching or exceeding the 3.5 MPa shear yield threshold in osteoporotic bone. Cortical segments near the metaphyseal flare experienced micromotion less than 5 μ m across all angulations, with interface shear below 2.0 MPa.

Construct stiffness in four-point bending measured 920 N/mm in Model A for the standard screw layout at 10 degrees angulation and 1.4 N·m torque, compared with 950 N/mm in bench tests on matched setups (3.2% difference). Force redistribution among screws showed that a 15-degree angulated distal screw carried 35% more axial load and experienced a 28% increase in bending moment relative to a coaxial screw under the same global load, consistent with strain gage measurements on screw shanks in surrogate tests (32% $\pm$ 6% increase).

Discretization error — gradation.

- a. Single mesh used; no quantitative study of mesh-induced

truncation.

- b. Two-mesh comparison on a subset of QoIs; qualitative judgment of adequacy.
- c. Systematic refinement on all QoIs with estimated truncation for each; thresholds defined.
- d. Systematic refinement with Richardson extrapolation and uncertainty bounds on each QoI across local and global measures.

5. Model B: Generic Bone–Plate Construct with Homogenized Bone

The generic construct model used a parametric distal radius geometry based on an average of five CT-derived shapes, scaled to match the bench-tested surrogates. The bone was represented as a transversely isotropic continuum with axial modulus 8 GPa, radial and circumferential moduli 5 GPa, shear modulus 3 GPa, and Poisson ratios 0.33–0.38, with parameters tuned to match construct bending and axial stiffness from bench data. A cortical shell region 1.5 mm thick with 14 GPa stiffness was included in the metaphysis to improve load pathway accuracy. The plate and screws matched the device geometry, but the conical seat was simplified with a filleted transition at the rim and a slightly smoothed cone–countersink junction to reduce mesh singularity.

The mesh comprised 0.9 million ten-node tetrahedra for the bone and 0.4 million for the plate and screws, with a 0.20 mm element size at the conical seat region and 0.35 mm elsewhere in the plate head; screw heads used 0.3 mm elements. The screw–bone interaction employed an elastic foundation distributed over the projected thread envelope with a stiffness of 180 MPa/mm in shear and 300 MPa/mm in compression, and a cap at 3.5 MPa shear to represent interfacial yielding in osteoporotic analogs. The head–seat contact used $\mu = 0.12$ as in Model A.

Under a 1.4 N·m seating torque at 15 degrees, Model B predicted peak von Mises at the seat rim of 820 MPa, with no plastic strain because the peak remained below the 900 MPa yield point. At 20 degrees and 2.0 N·m torque, peak von Mises reached 950 MPa with a small plastic zone of 0.1% equivalent strain, localized over a 0.15 mm arc length near the eccentric rim. Pressure–film measurements indicated 1,600–1,900 MPa contact pressure peaks at 15 degrees and 1.4 N·m, and the submodel (Model C) predicted a higher peak von Mises; we use these to bound Model B’s under-resolution of seat stress.

Global construct stiffness under four-point bending was 980 N/mm for a two-row distal screw configuration at 10 degrees angulation and 1.4 N·m torque, within +3.2% of bench tests (950 N/mm). Screw load redistribution under angulation showed that, at 15 degrees, the angled distal screw took 25% more axial load than a coaxial neighbor, consistent in direction and slightly lower in magnitude than Model A and bench strain gage data. For Mechanism 2, the homogenized interface shear envelope reached 2.1 MPa at 15 degrees and 2.9 MPa at 20 degrees in low-density bone analogs, with micromotion proxies from the

foundation model around 30 μm and 46 μm , respectively. The model captured the trend of increasing demand with angulation but, by construction, lacked trabecular-level micromotion fields. *Development process and product management — gradation.*

- a. Ad hoc development; limited documentation; unclear scope and configuration control.
- b. Basic planning and documentation; tracked changes but limited linkage to requirements.
- c. Structured planning with requirements traceability to models, tests, and QoIs; change control and reviews documented.
- d. Comprehensive planning with risk management, traceability matrices, and staged gates linking requirements to models and validation.

6. Model C: Plate–Screw Conical Seat Submodel (Motif)

The conical seat submodel extracted one plate hole with surrounding material in a 3 mm radius and the mating screw head, represented with full geometric fidelity of the conical interface and locking thread. The surrounding plate shell was extended to 1.2 times the seat diameter in thickness to capture bending constraint effects near the rim. A rotationally and translationally compliant connector represented the screw shank and threaded bone engagement, with stiffnesses calibrated against Model A's reaction wrench at the screw head for matched angulation and seating torque. The Ti-6Al-4V constitutive model included isotropic plasticity (yield 900 MPa) and a tangent modulus of 2 GPa up to 2% plastic strain, beyond which perfect plasticity was assumed as a conservative bounding.

Mesh density in the contact region was high: the smallest element edge length at the rim was 0.015 mm, with three h-refinement levels used in convergence studies (0.06, 0.03, 0.015 mm). The contact algorithm used a penalty approach with $\mu = 0.12$ baseline and 0.08–0.16 in sensitivity. Seating torque was converted to axial force through the conical head geometry and friction, but resolved directly via the solver to capture stick–slip transitions. For 1.4 N·m at 15 degrees, the peak von Mises at the rim was 990 MPa with a 0.2% plastic strain band approximately 0.1 mm wide; at 20 degrees and 2.0 N·m, the peak reached 1,360 MPa with 0.8% plastic strain and a local contact pressure maximum of 2,050 MPa. The contact map showed asymmetric engagement with a pressure-free sector opposite the eccentric load, as expected.

Sensitivity to friction showed that varying μ from 0.08 to 0.16 at 15 degrees and 1.4 N·m changed the peak von Mises by $\pm 6\%$ and the plastic zone length by ± 0.03 mm. At fixed $\mu = 0.12$, increasing seating torque from 1.0 to 2.0 N·m increased peak von Mises approximately linearly by 38%, from 980 MPa at 1.0 N·m to 1,350 MPa at 2.0 N·m at 15 degrees. While the model does not include the bone, it provides the head reaction wrench that informs screw shank bending; we used this to estimate head

lateral forces of 120 N at 15 degrees and 180 N at 20 degrees for a 250 N axial load. As a standalone, Model C is not intended to predict screw–bone micromotion.

Software quality assurance — gradation.

- a. Minimal controls; manual execution; undocumented environment.
- b. Source control and issue tracking in place; informal testing prior to runs.
- c. Version control with branching strategy; automated unit tests; defined build and run environments; peer review of changes.
- d. Comprehensive SQA with continuous integration, regression test suites, environment containerization, and release gates with documented artifacts.

7. Failure Mechanisms Governed by Screw Angulation

Two mechanisms organize the modeling and validation logic. The first is overstress at the plate-hole conical seat, where eccentric engagement at off-axis screw angles concentrates stress at the rim and can initiate permanent deformation or microcracks. This mechanism is sensitive to seating torque, friction, and local geometry at the countersink and rim. The second is degradation at the screw–bone interface, where off-axis screws generate increased shear and micromotion in trabecular regions and bending in the screw shank, promoting loosening or cutout. This mechanism is sensitive to bone quality, screw trajectory relative to cortical boundaries, and global load application.

Mechanisms interact: seat deformation can alter head alignment and, in turn, the load path into the screw, while increased head lateral force at high angulation changes the distribution of contact along bone threads. However, for the context-of-use and short-term early rehabilitation loads, we treat them separately: seat overstress is dominated by seating torque and the immediate head–seat engagement; screw–bone degradation is dominated by global loading and bone properties after seating is complete.

Results robustness — gradation.

- a. No sensitivity analysis; single value per input.
- b. One-at-a-time sensitivity for a few inputs; qualitative trend description.
- c. Structured sensitivity analysis covering key inputs; monotonicity and interaction checks; variance attribution.
- d. Comprehensive sensitivity and uncertainty quantification with sampling design, surrogate modeling, and variance-based decomposition.

8. Mechanism 1: Plate-hole conical seat overstress

Conical seat overstress arises from the asymmetric contact pattern that develops as the screw head axis tilts relative to the plate normal. At small angles, the contact band shifts but remains largely continuous; at larger angles and higher torques, the contact localizes into an arc sector, raising local pressures and stresses at the rim. We quantified the rim von Mises and principal stresses, plastic strain, and the distribution of contact pressure as functions of angulation and seating torque. In the submodel, peak stresses exceeded the Ti-6Al-4V yield point at 15 degrees with 1.4 N·m seating and at lower angles for 2.0 N·m seating. The patient-specific model reproduced this transition at slightly lower magnitudes due to assembly compliance and shank bending, whereas the generic construct, with a smoother conical transition, underpredicted peaks and produced limited plasticity except at the most severe settings.

Pressure-film mapping in bench seats at 15 degrees and 1.4 N·m displayed a brightening of the arc on the eccentric side, with calibrated peak pressures of 1,600–1,900 MPa and a contact-free region spanning about 90 degrees opposite. DIC measurements of plate-hole rim deformation after seating recorded 15–30 μm ovalization in hard torques near 2.0 N·m at 15 degrees, while FE predicted 18–25 μm in Model C, 9–12 μm in Model A, and 6–10 μm in Model B, consistent with differences in local compliance capture. The consistency of the shape and arc location across models supports the use of Model C to establish local limits and of Model A to assess seat response within the assembly when load sharing affects contact conditions.

Numerical code verification — gradation.

- a. No formal verification activities reported; reliance on software defaults.
- b. Basic benchmark problems solved; qualitative agreement.
- c. Quantitative comparison to analytical or highly resolved references with reported error norms on representative physics (elasticity, contact).
- d. Formal suite of benchmarks including manufactured solutions and contact-specific tests; error norms documented against acceptance thresholds.

9. Mechanism 2: Screw–bone interface degradation

Off-axis screw trajectories increase the lateral component of head reaction, elevating shank bending moments and altering the distribution of relative motion and shear along the threaded engagement. In trabecular regions, the increased shear can approach or exceed interfacial yield limits when bone is osteoporotic. In cortical regions, the interface tends to remain below slip thresholds, but local peaks can occur near transitional zones, especially where screw trajectories move close to cortical boundaries. We quantified the micromotion in trabecular and cortical windows and the shear stress envelope along the threads

as functions of angulation and loading. The patient-specific model provided detailed fields and localized peaks, whereas the homogenized model provided envelope predictions and smooth trends. The seat submodel, by design, does not include the bone and cannot provide these quantities directly.

Bench tests in polyurethane foam blocks (density 0.32 g/cm³) with acrylic cortical shells recorded 60 μm micromotion at 100 N eccentric axial load for 15-degree screws, while Model A predicted 55 μm and Model B predicted 48 μm under matched conditions. In combined loading at 250 N axial and 3 N·m bending, Model A predicted the trabecular micromotion rise from 22 μm to 45 μm as angulation increased from 0 to 15 degrees, and to 70 μm at 20 degrees, aligning with the trend in sawbone tests. Interfacial shear rose proportionally, approaching the 3.5 MPa envelope for low-density conditions. These results inform angulation limits in compromised bone and support the use of Model A and B for Mechanism 2 assessments.

Validation activities relevance — gradation.

- a. Measurements exist but are dissimilar to the computational scenario.
- b. Measurements partially overlap key aspects (geometry, loads, or installation).
- c. Measurements closely match primary aspects and cover most QoIs.
- d. Measurements replicate the computational scenario across all primary QoIs and boundary conditions.

10. Finite Element Implementation and Boundary Conditions (Implicit Solver)

All models were implemented in a commercial implicit finite element package with nonlinear contact and large-deformation capabilities. Solution controls enabled full Newton–Raphson with line search and automatic step size control. Contact stabilization used a small normal compliance corresponding to 0.1% of the characteristic stiffness of the contacting surfaces, and tangential behavior was governed by a Coulomb model with μ in the 0.08–0.16 range per sensitivity. Convergence tolerances were set to achieve residual force norms below 106 of the characteristic load and displacement corrections below 107 m in the most refined regions.

Seating simulations applied a torque boundary condition to the screw head via a distributed kinematic coupling, increasing from zero to target value over a pseudo-time step count of 20–50 increments, with automatic subincrementation as needed. Equilibrium was ensured at each increment before proceeding. Post-seating load steps applied axial force and bending moment via rigid surfaces, with combined loads resolved in 30–60 increments. For models A and B, the screw shank engagement reacted load via cohesive or foundation formulations; for Model C, the connector stiffness matched the head-level reaction compliance observed in the other models. Output extraction

scripts captured QoIs at the last converged increment of each step.

Numerical solver error — gradation.

- a. No quantified residual tracking; solver termination by default criteria.
- b. Residuals tracked; acceptance based on experience without explicit thresholds.
- c. Residuals and penetrations compared to predefined thresholds tied to QoIs; logs archived.
- d. Residuals, penetrations, and energy balance tracked and bounded with repeatability established across reruns and minor perturbations.

11. Verification Activities

Verification addressed two aims: correctness of the numerical implementation on representative physics and quantification of discretization effects on QoIs. For the former, we solved linear elastic patch tests on distorted quadratic tetrahedral meshes and achieved relative strain errors below 0.1%. We also validated the contact algorithm via a sphere-on-cone benchmark with known normal pressure distribution, obtaining peak pressure differences below 1.5% against a semi-analytical reference for $\mu = 0$ and below 3% for $\mu = 0.12$. An in-house manufactured-solution problem with spatially varying displacement and pressure fields in a pseudo-contact setting produced L2 norm errors under 0.2% at the fine mesh levels used in the production runs.

For discretization quantification, we executed h-refinement studies on each model targeting their key QoIs. Model C used three successively refined meshes (0.06, 0.03, 0.015 mm at the rim), yielding peak von Mises changes of 7% and then 2% across refinements at 15 degrees and 1.4 N·m; Richardson extrapolation produced an estimated truncation uncertainty of 3% on the peak stress. Model A used 0.20 and 0.10 mm local seat meshes, producing peak rim stress changes of 11% at 15 degrees and 1.4 N·m and 4% at 20 degrees and 2.0 N·m; trabecular micromotion convergence in the low-density region improved from 12% change at 1.2 to 0.8 mm element size to 4% change at 0.8 to 0.6 mm. Model B seat refinements from 0.40 to 0.20 mm changed peak rim stress by 18% at 15 degrees and 1.4 N·m, indicating under-resolution relative to Model C's standard; for Mechanism 2 envelopes, micromotion proxies changed by 7% from 0.35 to 0.25 mm in the thread envelope region.

Numerical code verification — gradation.

- a. No dedicated tests; reliance on vendor claims only.
- b. Basic elasticity and contact checks; visual inspection of fields.
- c. Quantified tests with error norms on elasticity and contact against known solutions; acceptance thresholds applied.

d.

Expanded suite including manufactured solutions and complex contact; error norms within tight thresholds with version-to-version regression tracked.

12. Numerical Code Verification

We applied a series of unit and integration tests at the solver and model-preprocessor interface. A 3D elasticity manufactured solution with polynomial displacement fields was imposed on a 40,000-element tet mesh, and computed strains and stresses matched the analytical fields within 0.18% in the L2 norm. The penalty contact algorithm was exercised with flat-on-cone and ring-on-cone setups across $\mu = 0$ –0.2, recovering expected normal pressure distributions with less than 1.5% peak deviation at $\mu = 0$ and less than 3.1% at $\mu = 0.12$ under coarse-to-fine transitions. Thread contact in the screw head locking feature reproduced axial stiffness within 2% of closed-form predictions from collar friction equations for the given μ .

Model assembly scripts were checked by cross-validating mass, stiffness, and contact area against CAD references to within 0.5% (mass) and 1.2% (contact area). These tests were tied to Git commit hooks and executed in a continuous integration pipeline on each revision affecting meshing, contact definitions, or material cards, with log retention to permit audit.

Software quality assurance — gradation.

- a. Manual builds and execution; limited test repeatability.
- b. Scripted builds; unit tests for utilities; no regression tracking.
- c. Automated builds with unit and integration tests; regression tracking for key metrics; code reviews required.
- d. Continuous integration with coverage metrics, static analysis, and signed releases; issue tracking integrated with test evidence.

13. Numerical Solver Error Characterization

Solver performance and residual control were monitored across all runs. For Models A–C, the final residual force norm was below 1.0e6 of the characteristic applied load in each step. Contact penetration at the conical seat was below 0.3% of the local element size at convergence (e.g., below 0.00045 mm for 0.15 mm elements and below 0.000045 mm for 0.015 mm elements in Model C). Energy balance deviations were under 0.5% in seating steps and under 0.3% in load application steps. Automatic step cuts occurred in fewer than five increments per step for seating and in fewer than three per step for loading, indicating stable convergence behavior.

Repeat runs with slightly perturbed initial conditions ($\pm 2\%$ change in initial torque ramp rate) yielded identical QoIs within 0.2% for Models A and C and within 0.5% for Model B, providing evidence of solver repeatability. For friction

sensitivity, convergence rates remained comparable across $\mu = 0.08\text{--}0.16$, although additional subincrementation was required near the onset of local slip transitions; QoIs remained within the range described in sensitivity sections.

Results robustness — gradation.

- a. Single-run outputs reported; robustness not investigated.
- b. Parametric sweeps for select parameters; trends qualitatively consistent.
- c. Systematic sweeps for key parameters with monotonicity checks and response surfaces for QoIs; robustness claims quantified.
- d. Global sensitivity with design-of-experiments sampling; variance-based attribution and uncertainty bounds reported.

$$CI_{\text{score}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\text{varepsilon}_{\text{rel}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\text{val}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

14. Discretization Error Estimation and Mesh Convergence

We designed mesh convergence studies around quantities that are sensitive to local gradients and contact singularities. In Model C, the rim stress and plastic zone length are strongly mesh-dependent near the countersink edge. The triplet of meshes (0.06, 0.03, 0.015 mm) demonstrated asymptotic behavior with a 7% reduction in peak von Mises from coarse to medium and a further 2% from medium to fine at 15 degrees and 1.4 N·m. Applying a Richardson extrapolation with an assumed order of 1.8 (consistent with quadratic tetrahedra in contact) yielded an estimated fine-mesh peak of 1,020 MPa and a truncation uncertainty of 3%. The selected fine mesh is therefore expected to slightly underpredict the true peak by approximately 3% at this setting.

In Model A, local refinement at the seat from 0.20 to 0.10 mm reduced peak rim stress by 11% at 15 degrees and 1.4 N·m, and 4% at 20 degrees and 2.0 N·m. Micromotion in trabecular regions converged as the trabecular core mesh refined from 1.2 to 0.8 to 0.6 mm: the 15-degree, 250 N axial plus 3 N·m bending case changed from 49 to 46 to 45 μm , implying a 4% remaining truncation at 0.6 mm. In Model B, seat refinement from 0.40 to 0.20 mm reduced the 15-degree, 1.4 N·m rim stress from 900 to 820 MPa (8.9%), while Mechanism 2 envelope micromotion proxies changed from 32 to 30 μm at 15 degrees (6.3%) when refining the thread envelope mesh. These studies set the basis for factor gradations related to mesh-induced truncation and identify Model B's seat under-resolution as a target for elevation.

Discretization error — gradation.

- a. No quantitative mesh study; mesh selection justified qualitatively.
- b. Limited mesh refinement on a subset of the domain; local QoIs assessed.
- c. Systematic refinement for all QoIs with estimated truncation; mesh chosen to meet defined thresholds.
- d. Systematic refinement with extrapolation and error bars carried into validation and decision-making.

15. Model Form Rationale, Assumptions, and Idealizations

Model choices reflected the mechanisms of interest and the context-of-use. In Model A, the heterogeneous bone model, explicit screw head–seat geometry, and cohesive-like screw–bone representation enabled simultaneous assessment of both seat overstress and screw–bone micromotion. The assumption of isotropic hardening for Ti-6Al-4V captures the limited plasticity expected at the rim. Bone was modeled as linear elastic to isolate interface phenomena under early rehabilitation loads; viscoelasticity was neglected as a second-order effect. In Model B, the homogenized, transversely isotropic bone provided a construct-level stiffness match but necessarily smoothed local fields; the conical seat simplification reduced stress concentration fidelity but promoted mesh stability. In Model C, the submodel idealizes the screw shank as springs that match global compliance, allowing concentration of degrees of freedom in the contact region to resolve seat stress and plasticity with high fidelity.

Idealizations include neglecting thread-form engagement in the bone in Model B (replaced by an elastic foundation) and smearing thread engagement in Model A via cohesive behavior rather than resolving each flank in trabecular bone. The conical seat rim geometry was matched to CAD in all models, but the plate surface finish and microfeatures were not modeled. Friction coefficients were set based on literature and supplier data for titanium–titanium and titanium–bone interfaces and explored in sensitivity. The models were not intended to simulate late biological responses such as bone remodeling; the time window is the early postoperative period where mechanics dominate.

Model form — gradation.

- a. Empirical correlation with limited mechanistic basis; not traceable to physical phenomena.
- b. Mechanistic representation with notable simplifications in geometry or material behavior.
- c. Mechanistic representation with justified simplifications; primary physics resolved for QoIs.
- d. High-fidelity mechanistic representation with all primary physics and key secondary effects resolved for QoIs.

16. Validation Activities and Their Relevance to the Assessment

We executed four classes of bench tests. First, instrumented seating tests tightened conical-head screws into isolated plate seats while measuring applied torque and resulting axial force; Fuji Prescale films recorded contact pressure maps at 0, 10, and 15-degree angulations and 1.0–2.0 N·m torques. Second, plate-hole deformation was measured in situ using DIC at 10× magnification, capturing ovalization after seating. Third, distal radius constructs with plate and screws were tested in axial compression and four-point bending under loads matching model inputs, with screw shank strain gages to quantify load redistribution when angulated. Fourth, screw toggle and micromotion were measured in foam and cadaveric surrogates under eccentric loads with reflective markers at the screw head and a fluoroscopic motion-tracking system providing micromotion in trabecular analogs. Each test class was repeated across angulations and torques to create response surfaces for model comparison.

Comparisons between models and measurements included: for seat pressure maps, Model C predicted peak pressure 1,850 MPa at 15 degrees and 1.4 N·m, within the 1,600–1,900 MPa film range; Model B predicted lower peaks (1,500–1,700 MPa equivalent) in line with smoother geometry. For rim ovalization, DIC reported 15–30 µm at 2.0 N·m and 15 degrees, while Model C predicted 25 µm and Model A 12 µm. Construct stiffness in bending was 950 N/mm experimentally; Model A predicted 920 N/mm (3.2%) and Model B 980 N/mm (+3.2%). Micromotion at 100 N eccentric load in foam blocks at 15 degrees was measured at 60 µm; Model A predicted 55 µm and Model B 48 µm. These comparisons guided factor gradations regarding measurement match and the directness of applicability.

Relevance of the validation activities to the COU — gradation.

- a. Measurements differ substantively from the modeled scenarios.
- b. Measurements match on some aspects (load, installation, geometry) but require extrapolation for others.
- c. Measurements match on most aspects, with minor extrapolation.
- d. Measurements replicate the modeled scenarios across the primary QoIs.

17. Software Quality Assurance and Development Management

Configuration control and reproducibility were implemented via a Git repository with protected main branch, pull requests, and code reviews required for all changes affecting meshing, materials, contact, or analysis scripts. Continuous integration executed 68 unit tests and 14 integration tests spanning mesh generation, contact pair definitions, material card assignment,

and QoI extraction. Static analysis checked template consistency and boundary condition completeness. Releases were versioned; v1.6.2 was used for all results herein. Execution environments were containerized to lock solver version, libraries, and hardware dependencies. Artifacts (meshes, input decks, logs) were archived with metadata for traceability.

Planning and oversight of model development followed a stage-gate process: scoping and requirements definition, verification planning, validation planning, implementation, and release. Requirements traced to named QoIs and to validation tests in matrices maintained alongside the repository, linking model changes to test coverage. Risk registers identified potential failure modes in modeling (e.g., under-resolved seat rim) and planned mitigations (e.g., submodeling). While we conducted internal cross-functional meetings to discuss progress and risks, we do not present a formal report of a development technical review herein. Prior uses of similar modeling patterns in other plate systems informed early assumptions, but we provide no consolidated use history summary in this paper.

Development process and product management — gradation.

- a. Informal planning; limited traceability and risk tracking.
- b. Documented plans with partial traceability and ad hoc risk discussions.
- c. Structured plans with trace matrices linking QoIs, models, and tests; risk registers maintained.
- d. Comprehensive product management with staged reviews, traceability from requirements to validation, and risk mitigation plans executed.

18. Engineering Reviews and Use History (Summary)

Interim design and analysis reviews were held at key milestones to exchange findings among modeling, test, and design engineers. Review notes captured action items such as densifying the Model B seat mesh and adding a 20-degree angulation case to the validation plan. The tone and scope were consistent with internal engineering assessments rather than formal technical reviews with external independent oversight.

Elements of the modeling toolkit have been applied previously to volar plates of different geometries and to proximal humerus plates; these experiences informed choices in contact and cohesive parameters. We do not present a curated analysis of prior performance or a quantitative use history across devices, as the focus here is the distal radius plate family and two mechanisms under the present context-of-use.

Development technical review — gradation.

- a. No structured reviews; informal discussions only.
- b. Periodic internal reviews focused on progress; limited documentation of decisions.

- c. Structured internal reviews with documented decisions and action tracking; selective external input.
- d. Independent formal technical reviews with closure of findings and documented impact on plans.

19. Results Robustness: Parameter and Boundary-Condition Sensitivity

We explored sensitivities to angulation (0, 10, 15, 20 degrees), seating torque (1.0, 1.4, 2.0 N·m), friction μ (0.08, 0.12, 0.16), and bone density (0.25, 0.40, 0.80 g/cm³). In Model C, peak rim von Mises at 15 degrees rose linearly with torque from 980 MPa (1.0 N·m) to 1,350 MPa (2.0 N·m), and varied $\pm 6\%$ with μ from 0.08 to 0.16. In Model A, the trabecular micromotion at the distal screw under 250 N axial plus 3 N·m bending increased from 22 μ m (0 degrees) to 45 μ m (15 degrees) and 70 μ m (20 degrees) in low-density bone; in normal-density bone (0.80 g/cm³), the same sequence yielded 14, 28, and 41 μ m. In Model B, the screw load redistribution at 15 degrees and 1.4 N·m showed a 25% axial load increase, changing by $\pm 4\%$ across friction extremes and $\pm 6\%$ across bone density levels.

A Latin hypercube sampling (LHS) of 120 runs in Model A spanned angulation 0–20 degrees, torque 1.0–2.0 N·m, μ 0.08–0.16, and ρ 0.25–0.80 g/cm³. Linear regression with interaction terms explained 89% of the variance in rim stress and 92% in micromotion, with angulation and torque as the dominant contributors to rim stress (standardized coefficients 0.61 and 0.47) and angulation and bone density dominating micromotion (0.58 and 0.53). No sign reversals occurred in response surfaces across the ranges tested. Similar sweeps in Model B explained 85% of variance in envelope shear and 88% in stiffness. These findings indicate that input–output relations remained monotonic and that key outputs were controlled primarily by the intended parameters.

Results robustness — gradation.

- a. Limited or no sensitivity checks; results could be fragile.
- b. Sensitivity on a subset of inputs; monotonicity partially established.
- c. Systematic sensitivity on key inputs; monotonicity and variance attribution demonstrated.
- d. Comprehensive robustness with design-of-experiments and uncertainty quantification supporting decision confidence.

20. Credibility Scoring Methods and Rubric (0–7 Index)

We graded each factor on a 0–7 internal index, where 0 indicates absent evidence or inapplicability and 7 indicates the highest level of rigor within this assessment framework.

The scale aligns conceptually with the gradations in NASA-STD-7009A but is presented on a private 0–7 index for clarity and aggregation. Scores were assigned independently for each model-by-mechanism pairing. The basis for each score draws on verification and validation results, sensitivity analyses, and process documentation as described in the body prose. We also computed an overall index per model-by-mechanism as the arithmetic mean across the eight factors, used internally to prioritize elevation work; we report these means descriptively in the synthesis section without supplanting factor-wise interpretation.

To promote transparency, we include ladder-style rubrics that articulate increasing rigor for key factors. These ladders informed how we mapped our evidence to the 0–7 scale. Our tables present the factor, the model and mechanism scope, the numeric level only in the LEVEL column, and a one-line basis that restates findings and cites numbers also present in the prose. *Credibility scoring and aggregation — gradation.*

- a. Factor scores assigned qualitatively without explicit criteria; aggregation opaque.
- b. Criteria described; scores assigned with examples; aggregation described qualitatively.
- c. Structured rubrics tied to evidence; scores traceable; mean reported with context.
- d. Structured rubrics and calibration across scorers; scores traceable; aggregation with uncertainty bounds and sensitivity to weighting.

21. Credibility Scorecards by Model $\times$ Mechanism: Model A

Model A addresses both mechanisms with detailed assembly fidelity and heterogeneity. For Mechanism 1, mesh refinement bounded rim stress truncation to approximately 4% on the selected mesh, solver residuals were below 1e6 with contact penetrations under 0.3% of element size, and validation against pressure-film and DIC yielded shape and magnitude matches within 10–20%. For Mechanism 2, micromotion and interface shear matched foam and surrogate measurements within 8–20%, with monotonic trends across angulation and density. Process controls, SQA, and code verification were uniformly strong across both mechanisms.

The table below lists factor-by-factor gradations and concise bases restating key quantitative evidence.

22. Credibility Scorecards by Model $\times$ Mechanism: Model B

Model B emphasizes construct-level stiffness and load sharing with a homogenized bone representation. For Mechanism 1, simplifications at the conical seat reduce the fidelity of peak stress prediction; for Mechanism 2, envelope predictions track measurements and Model A trends with acceptable error.

Solver behavior and code verification apply uniformly, and SQA practices are consistent across models.

The table below presents factor gradations and numerical bases for each mechanism.

23. Credibility Scorecards by Model × Mechanism: Model C

Model C is a high-fidelity local submodel optimized for Mechanism 1 and used in conjunction with head reaction wrenches from the assembly models. It delivers converged local seat stress, plasticity, and pressure distributions with explicit representation of contact and plasticity near the rim. It is not intended, on its own, to predict bone micromotion or interfacial shear; we assessed its Mechanism 2 credibility accordingly. Solver behavior and code verification match the other models, and SQA practices are uniform.

The table summarizes factor gradations and quantitative bases for each mechanism.

24. Mechanism 1: Per-Model Factor Scores (0–7)

Model A: For seat overstress, the selected local mesh at the rim (0.10 mm) leaves approximately 4% truncation uncertainty based on the 11% change observed from 0.20 to 0.10 mm at 15 degrees and 1.4 N·m and the 4% change at 20 degrees and 2.0 N·m. Solver residuals were consistently below 106, and contact penetrations stayed under 0.3% of the local element size, with energy balance drift below 0.5% and fewer than five cutbacks per step. Model form captured elastic–plastic behavior at the rim with 0.3% equivalent plastic strain at 15 degrees and 1.4 N·m and 0.6% at 20 degrees and 2.0 N·m. Validation matched film peaks (1,600–1,900 MPa) within 10% and DIC ovalization (15–30 μ m) with a prediction of 12 μ m, which is conservative due to assembly compliance. Sensitivity to torque showed near-linear scaling of peak stress from 980 MPa at 1.0 N·m to 1,350 MPa at 2.0 N·m, and friction μ from 0.08 to 0.16 changed peaks by approximately $\pm 6\%$.

Model B: For seat overstress, the smoothed seat geometry underpredicted peak rim von Mises stress, with 820 MPa at 15 degrees and 1.4 N·m compared with 990 MPa in the submodel, and allowed only 0.1% plastic strain at 20 degrees and 2.0 N·m. Refinement at the rim from 0.40 to 0.20 mm reduced peaks by 8.9%, indicating that truncation remains a material contributor to error. Solver metrics matched those of Model A, with residuals under 106 and stable increment control. Validation against film data remained within 12% of the lower bound of measured peaks, consistent with geometric smoothing. Sensitivity analyses demonstrated monotonic increases with torque and angulation; friction variation induced approximately $\pm 5\%$ change in peaks.

Model C: For seat overstress, the triple refinement sequence (0.06, 0.03, 0.015 mm) reduced peak von Mises by 7% then 2% at 15 degrees and 1.4 N·m, with Richardson extrapolation suggesting a 3% residual truncation. The model reproduced DIC ovalization at 25 μ m for 15 degrees and 2.0 N·m within the 15–30 μ m measured range and predicted pressure film peaks

of 1,850 MPa at 15 degrees and 1.4 N·m. Solver behavior was excellent: residuals below 106, penetrations under 0.3% of 0.015 mm elements, and energy drift below 0.5%. Parametric sensitivity with torque and friction preserved monotonicity and produced $\pm 6\%$ peak variations with μ .

25. Mechanism 2: Per-Model Factor Scores (0–7)

Model A: The cohesive interface model and heterogeneous bone captured micromotion and shear trends with angulation. At 250 N axial plus 3 N·m bending, trabecular micromotion at the distal-most screw increased from 22 μ m at 0 degrees to 45 μ m at 15 degrees and 70 μ m at 20 degrees for low-density regions (0.25 g/cm³), with cortical segments under 5 μ m. Shear rose from 1.2 to 2.8 to 3.6 MPa across those angles, reaching the 3.5 MPa threshold near 20 degrees. Foam tests at 100 N eccentric load measured 60 μ m at 15 degrees; the model predicted 55 μ m (8%). Construct stiffness was 3.2% versus bench, and screw load increases of 35% in axial and 28% in bending matched gage trends. Mesh convergence for micromotion was adequate with a 4% change from 0.8 to 0.6 mm trabecular elements. Solver performance and repeatability matched Mechanism 1.

Model B: The elastic foundation captured micromotion envelopes and stiffness but not trabecular detail. At 15 degrees and 1.4 N·m under 100 N eccentric load, micromotion proxy was 48 μ m (20% vs. 60 μ m measured). Under 250 N axial plus 3 N·m bending, the shear envelope reached 2.1 MPa at 15 degrees and 2.9 MPa at 20 degrees, with micromotion proxies of 30 and 46 μ m, respectively. Load redistribution was +25% in axial load for the angled screw, slightly conservative compared with Model A and gage data. Mesh refinement of the thread envelope changed micromotion proxies by 6.3%, acceptable for envelopes. Solver performance and validation relevance were strong for this mechanism.

Model C: The submodel lacks bone and cannot compute micromotion or shear. We used head reaction as a proxy, which increased from 120 N lateral at 15 degrees to 180 N at 20 degrees under 250 N axial load. However, refining the submodel (0.06 to 0.015 mm) changed the proxy lateral force by 22% at 20 degrees and 2.0 N·m, indicating that this quantity is not a robust stand-in for interface behavior. As such, credibility for Mechanism 2 remains low for model form, discretization on the proxy QoI, and validation relevance, despite solver and code verification strength.

26. Synthesis of Findings and Elevation Priorities

Trace matrices linked user needs, physics targets, and tests; coverage exceeded 95% of the named requirements across the three models, and gating ensured that each planned quantity of interest was mapped to at least one validation activity or benchmark prior to release.

Mesh refinement studies bounded numerical truncation to under 4% in the reported quantities of interest across all three models at their selected mesh densities, with the exception of the seat region in Model B, which we flagged for elevation.

The chosen constitutive descriptions and geometric detail reproduced observed response trends and magnitudes within the intended envelope, with the primary approximation residing in the smoothed seat geometry of Model B, which biased peak stress predictions low relative to Model C references.

Independent benchmarks and manufactured-solution tests produced deviations under 0.2% for elasticity and under 3.1% for contact peaks, supporting the correctness of the numerical implementation for the physics encountered.

Equilibrium and contact metrics remained strictly controlled: the residual force norm was under 106 of the characteristic load, and penetrations at contact interfaces were limited to less than 0.3% of the local element size across all runs.

The bench protocols matched angulation, torque, and loading envelopes within 10% of the intended use, and measurement modalities (film, DIC, gages, motion tracking) overlapped the modeled outputs so that comparisons were direct for the principal quantities of interest.

Across torque, friction, angulation, and bone density sweeps, the direction and magnitude of change in outputs were preserved, with 85–92% of response variance attributable to the intended controls in Latin hypercube and one-at-a-time analyses.

Version control, automated testing, and review gates were exercised on every change set, with 68 unit tests and 14 integrations passing on release v1.6.2, and all production runs executed in containerized environments to ensure reproducibility.

Priorities to elevate credibility include: (1) refining the seat geometry and mesh in Model B to reduce peak stress underprediction for Mechanism 1; (2) extending Model C with a thread-engagement layer or coupling it to a local bone patch to enable limited Mechanism 2 assessments; and (3) expanding the patient cohort for Model A to three to five additional anatomies to strengthen generalizability while maintaining current fidelity. We also plan to widen torque and friction sampling in validation to include highly lubricated and roughened seats to test the bounds of friction sensitivity.

Aggregation and decision-making — gradation.

- a. Narrative synthesis without clear linkage to factors or planned actions.
- b. Synthesis cites factor scores; actions described qualitatively.
- c. Synthesis ties scores to quantified gaps and prioritized actions with measurable targets.
- d. Synthesis formalizes action plans with acceptance criteria, timelines, and re-assessment triggers.

27. Limitations and Planned Extensions

The models focus on early postoperative mechanical behavior without biological adaptation or time-dependent changes in bone or interface properties. Viscoelastic and plastic behavior in trabecular bone was neglected, which could affect micromotion

under sustained loads. Friction was modeled as Coulomb with a constant coefficient; potential variations due to surface treatments, wear-in during seating, or lubrication changes were only bracketed in sensitivity. Manufacturing tolerances and surface finish in the conical seat and screw head were idealized. Model A represents a single anatomy; while it reproduces the bench fixtures used for validation, anatomical variability can alter local boundary conditions near the plate head.

Planned extensions include: (1) adding two to four additional patient-specific assemblies to broaden anatomical coverage; (2) implementing an enriched conical seat representation in Model B to better resolve local peaks without sacrificing computational efficiency; (3) coupling Model C to a local trabecular patch with a thread-form-inspired interface to permit limited Mechanism 2 checks; (4) expanding validation to include higher-resolution DIC around the plate rim and in situ seat pressure mapping using embedded sensors; and (5) broadening parameter sampling to include roughened versus polished seats and lubricated conditions to probe friction-dependent uncertainties.

Use history — gradation.

- a. No prior applications documented.
- b. Similar models applied in related devices with anecdotal outcomes.
- c. Documented prior applications with qualitative performance summaries.
- d. Documented prior applications with quantitative performance and relevance to current context.

28. Conclusions

This assessment applied a structured, factor-by-factor approach to evaluate three implicit finite element models for analyzing the effect of screw angulation in volar locking distal radius plates under early postoperative loads. Model C best established local limits for conical seat stress and plasticity with fine control of contact and material nonlinearity; Model A bridged local and global phenomena by resolving seat behavior within an anatomical assembly and directly predicting screw–bone micromotion and shear; Model B efficiently captured construct stiffness and load redistribution while providing envelope predictions for interface behavior. Bench validation aligned with the context-of-use for both mechanisms, and verification and solver controls provided confidence in the numerical foundations.

For Mechanism 1, the combination of Model C and Model A supports design decisions on allowable angular ranges and seating torque recommendations, with Model B reserved for construct-level what-if scenarios. For Mechanism 2, Model A is primary, with Model B as a useful adjunct for envelope trends; Model C is not applicable in isolation. Credibility gradations and their evidence, reported as 48 entries across three tables, enable transparent mapping from evidence to use.

Elevation work identified here—seat resolution in Model B, limited Mechanism 2 capacity in Model C, and broader anatomic sampling in Model A—will further strengthen decision support in subsequent cycles.

29. Supplemental Material

Supplemental files include: (1) mesh convergence plots for each model and QoI; (2) seating torque versus axial force curves for instrumented conical seat tests at 0, 10, and 15 degrees; (3) DIC maps of plate-hole rim deformation at 15 and 20 degrees under 1.4 and 2.0 N-m seating; (4) raw pressure-film images and calibration curves; (5) screw shank strain gage time histories under four-point bending; (6) Latin hypercube design matrix and regression coefficients for Models A and B; and (7) container recipes and run scripts for reproduction with solver version and library hashes.

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Table 1: Model A credibility scorecard across Mechanism 1 (plate-hole conical seat overstress) and Mechanism 2 (screw–bone interface degradation).

Credibility factor	Level	Basis
Development process and product management - Model A - Mechanism 1	6	Stage-gate plan with requirements-to-QoI trace; >95% requirements linked to tests; risk of seat under-resolution mitigated by submodeling.
Discretization error - Model A - Mechanism 1	5	Seat rim h-refinement 0.200.10 mm changed peak by 11% at 15°/1.4 N·m; estimated remaining truncation 4% at selected mesh.
Model form - Model A - Mechanism 1	5	Explicit seat geometry and Ti-6Al-4V plasticity captured rim yielding (0.3% strain at 15°/1.4 N·m); assembly compliance reduced peaks vs. sub-model.
Numerical code verification - Model A - Mechanism 1	6	Elasticity MMS L2 error 0.18%; contact benchmark peak error 1.5–3.1%; head-thread stiffness within 2% of collar-friction closed form.
Numerical solver error - Model A - Mechanism 1	6	Residual norm <1e6; contact penetration <0.3% of 0.10 mm; energy balance drift <0.5%; <5 cutbacks per step.
Relevance of the validation activities to the COU - Model A - Mechanism 1	6	Pressure-film at 0–15° and 1.0–2.0 N·m matched modeled angulation/torque; peaks within 10% of film range (1,600–1,900 MPa).
Results robustness - Model A - Mechanism 1	5	Peak rim stress scaled 9801,350 MPa with torque 1.02.0 N·m; ±6% with μ 0.08–0.16; monotonic trends confirmed in LHS.
Software quality assurance - Model A - Mechanism 1	6	Git CI with 68 unit and 14 integration tests; containerized environment; v1.6.2 release artifacts archived.
Development process and product management - Model A - Mechanism 2	6	Trace matrices linked micromotion QoIs to foam and cadaveric tests; planned LHS supported variance attribution.
Discretization error - Model A - Mechanism 2	5	Trabecular mesh 1.20.80.6 mm changed 15° micromotion 494645 μ m; 4% remaining truncation at 0.6 mm.
Model form - Model A - Mechanism 2	6	Heterogeneous bone and cohesive interface reproduced 60 μ m foam micromotion (predicted 55 μ m, 8%) and shear envelopes.
Numerical code verification - Model A - Mechanism 2	6	Same MMS and contact benchmarks; cohesive interface validated to pullout stiffness within 5% on foam analogs.
Numerical solver error - Model A - Mechanism 2	6	Residual norm <1e6; penetration <0.3% of element; repeat runs within 0.2% QoI variability.
Relevance of the val-	6	Foam and sawbone tests at 0–

Table 2: Model B credibility scorecard across Mechanism 1 (plate-hole conical seat overstress) and Mechanism 2 (screw–bone interface degradation).

Credibility factor	Level	Basis
Development process and product management - Model B - Mechanism 1	5	Planned scope focused on construct stiffness; risk of seat under-resolution acknowledged; elevation item logged to densify 0.20 mm rim mesh.
Discretization error - Model B - Mechanism 1	3	Seat rim 0.400.20 mm changed peak from 900820 MPa (8.9%); under-resolution relative to submodel indicates higher truncation.
Model form - Model B - Mechanism 1	3	Smoothed cone–countersink transition underpredicted rim peaks (820 MPa vs. 990 MPa in submodel at 15°/1.4 N-m) with limited plasticity.
Numerical code verification - Model B - Mechanism 1	6	Elasticity and contact benchmarks within 0.2–3.1% as for other models; same solver and preprocessor tests applied.
Numerical solver error - Model B - Mechanism 1	6	Residual norm <1e6; contact penetration <0.3% at 0.20 mm mesh; <3 cutbacks per load step.
Relevance of the validation activities to the COU - Model B - Mechanism 1	4	Seat pressure–film maps matched angulation/torque but geometry simplification reduced peak magnitude fidelity; within 12% of lower film bound.
Results robustness - Model B - Mechanism 1	4	Peak rim stress rose with torque and angle; μ 0.08–0.16 induced $\pm 5\%$ change; trends monotonic but peaks biased low.
Software quality assurance - Model B - Mechanism 1	6	Same Git CI and containerization; artifacts archived; version parity with Model A.
Development process and product management - Model B - Mechanism 2	5	Requirements targeted stiffness and load sharing; validation to sawbone bending within +3.2%; micromotion envelopes included.
Discretization error - Model B - Mechanism 2	4	Thread-envelope mesh 0.350.25 mm changed 15° micromotion proxy 3230 μ m (6.3%); acceptable for envelope metrics.
Model form - Model B - Mechanism 2	5	Homogenized bone with elastic foundation reproduced 60 μ m foam micromotion (48 μ m, 20%) and load redistribution (+25%).
Numerical code verification - Model B - Mechanism 2	6	Same MMS and contact benchmarks; foundation law checked against pullout stiffness surrogate within 6%.
Numerical solver error - Model B - Mechanism 2	6	Residual norm <1e6; repeatability within 0.5% on envelope QoIs; stable increments across angulation range.
Relevance of the validation activities to the COU - Model B - Mechanism 2	6	Construct bending and foam micromotion matched model loads and angles; errors 3–20% in stiffness and micromo-

Table 3: Model C credibility scorecard across Mechanism 1 (plate-hole conical seat overstress) and Mechanism 2 (screw–bone interface degradation).

Credibility factor	Level	Basis
Development process and product management - Model C - Mechanism 1	5	Submodel introduced per risk plan to resolve seat peaks; boundary wrench calibrated to Model A; scope defined in requirements.
Discretization error - Model C - Mechanism 1	6	Rim mesh 0.060.030.015 mm changed peak 7% then 2%; Richardson U 3% at 15°/1.4 N-m; plastic zone length change <0.03 mm.
Model form - Model C - Mechanism 1	6	Explicit seat geometry and Ti-6Al-4V plasticity reproduced DIC ovalization (25 μ m vs. 15–30 μ m) and film peak (1,850 MPa).
Numerical code verification - Model C - Mechanism 1	6	Contact and MMS benchmarks within 0.2–3.1%; head stiffness within 2% of collar-friction reference; logs archived.
Numerical solver error - Model C - Mechanism 1	6	Residual <1e6; penetration <0.3% of 0.015 mm; energy drift <0.5%; <5 cutbacks per seating step.
Relevance of the validation activities to the COU - Model C - Mechanism 1	5	Seat tests matched angles and torques; local seat-only context matched submodel; assembly effects assessed via boundary wrench.
Results robustness - Model C - Mechanism 1	5	Peak stress scaled 9801,350 MPa with torque; $\pm 6\%$ with μ ; trends preserved across 0–20°; no sign reversals.
Software quality assurance - Model C - Mechanism 1	6	Same CI, tests, and containerization; versioned artifacts; reproducible runs.
Development process and product management - Model C - Mechanism 2	5	Scope explicitly excludes bone micromotion; used to provide head wrench to other models per plan; limitations documented.
Discretization error - Model C - Mechanism 2	2	Proxy metric (head lateral force) changed 22% on refinement at 20°/2.0 N-m; no convergence for bone micromotion (not modeled).
Model form - Model C - Mechanism 2	2	No bone representation; cannot compute interface shear or micromotion; only head reaction available (120 N at 15°, 180 N at 20°).
Numerical code verification - Model C - Mechanism 2	6	Code checks as above; solver correctness established though QoIs for Mechanism 2 are out of scope.
Numerical solver error - Model C - Mechanism 2	6	Residual <1e6; repeatability within 0.2%; solver stable though QoIs are proxies.
Relevance of the validation activities to the COU - Model C - Mechanism 2	1	No direct measurements for micromotion or shear at seat-only scale; submodel not aligned to this mechanism.
Results robustness - Model C - Mechanism 2	2	Proxy head lateral force increased with angle but refinement sensitivity 22% at 20°